Echo Breakers: Phase 1 Decisions, Workflow and Plan

Oct 14, 2026 · @Trung Hieu Duong

Phase 1 requirements

Phase 1 asks for a PDF visual summary of at most 5 pages, in English, uploaded by 11:59 PM ET on Oct 18; the top 20 teams advance to Phase 2.

Item	Requirement
Task	Use publicly available datasets and other appropriate data sources to investigate patterns in online information exposure and recommendation systems
Deliverable	A data-driven visual summary: a graphic or series of graphics showing patterns, disparities and insights
Format	PDF, 5 pages maximum, English
Must include	A clear visual story, with data sources, methodology and limitations clearly identified
Deadline	Upload to the platform by 11:59 PM ET, Oct 18
Judging	A panel scores submissions Oct 19 to 21; teams get scores the evening of Oct 21
Outcome	The top 20 teams advance to Phase 2

The brief suggests exploring six themes, and each one has a place in our plan:

Theme from the brief	Where we cover it
How different users or communities are exposed to different types of information	MIND: shown vs clicked by category, behavior-based reader groups (page 3)

Theme from the brief	Where we cover it
Whether topics, perspectives, sources or geographic areas are over- or underrepresented	MIND categories (page 3), GDELT coverage by country (page 4)
How personalization may affect the diversity of information users encounter	Entropy of shown vs clicked content (page 3)
Potential indicators of bias or unequal outcomes in recommendation systems	Gini coefficient, catalog coverage, popularity share (pages 2 and 3)
How fairness and diversity in information exposure can be measured	The four metrics, explained in plain words (page 2)
Limitations or biases within the data itself	Limitations section (page 5)

Suggested datasets in the brief: news recommendation data, media and geographic datasets, and movie or content recommendation datasets; teams may also choose another relevant public dataset.

Decisions so far

We are building a news-exposure story on MIND-small, backed by a small GDELT slice for geography, and delivering it as a 5-page PDF by Oct 18.

Topic	Decision	Why
Team name	Echo Breakers	Speaks directly to filter bubbles and is easy to say in a presentation
Main dataset (A)	MIND-small, train set	Real impressions (shown) and clicks, with news categories; fits "information exposure" best
Supporting dataset (C)	GDELT, one month, top ~50 countries	Adds the geographic angle MIND lacks; keeps the news theme
Core metrics	Shannon entropy, Gini coefficient, catalog coverage, popularity share	Cover diversity, inequality of exposure and reach with simple, explainable numbers

Topic	Decision	Why
"Communities"	Behavior-based groups (for example by dominant category)	MIND has no demographics, so we say this openly in limitations
Deliverable	Visual summary, 5 pages max, PDF, English	Challenge requirement; judges score the visual story, sources, method and limitations
Tooling	Python (pandas, matplotlib) using the shared starter script `mind_analysis.py`	Same code for everyone; results are repeatable

How we work

Every task has one owner, one shared place for files, and a review before anything goes into the PDF.

Roles (names to be filled in once the team is confirmed; one person can hold two)

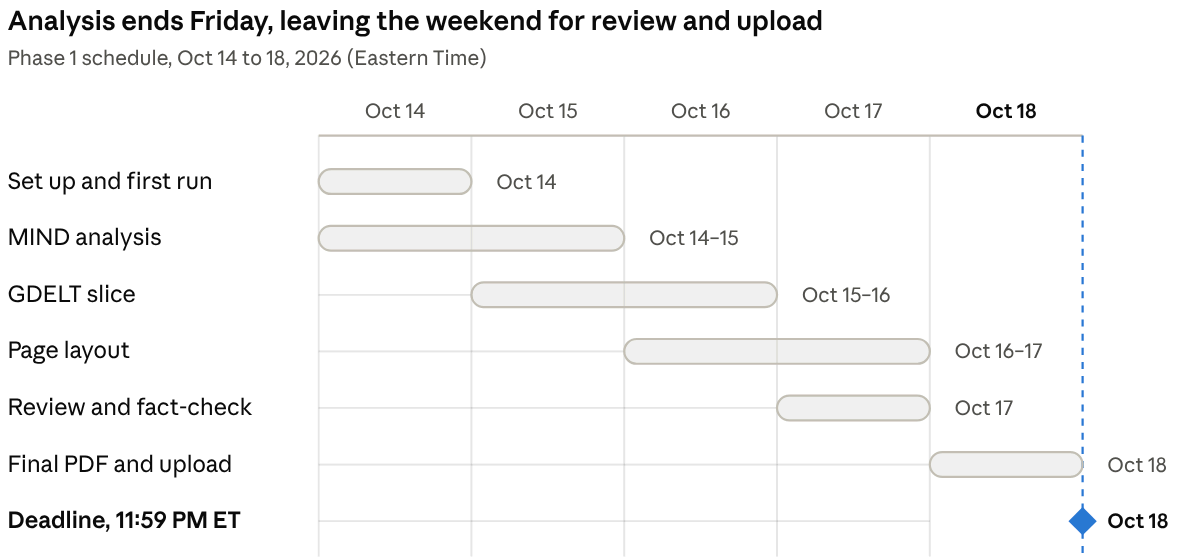
Role	Responsibility
Lead / project manager	Keeps the schedule, runs check-ins, makes the final call on scope
Data analyst (MIND)	Runs the analysis, owns the metrics and the numbers on every chart
Data analyst (GDELT)	Pulls and cleans the small geographic slice, owns the regional chart
Designer / storyteller	Lays out the 5 pages, keeps one visual style and a clear story
Reviewer / fact-checker	Checks every number against the output files and every claim against the limitations

Working rules

- Code and results live in one shared repository (GitHub); no analysis only on someone's laptop.

- Each chart is saved as a PNG with the script and the data file it came from, so any number can be traced.
- A chart goes into the PDF only after the reviewer has checked it.
- We write down every new decision in the Decisions table above, with the reason.
- Quick check-in once a day (15 minutes, same time) and one written update in the team chat at the end of each day: done, blocked, next.
- Scope changes go to the lead; if it cannot be finished by Friday, it is cut, not rushed.
- Sources and limitations are written as we go, not on the last night.

Plan and timeline



plan timeline · 6 steps, deadline Oct 18

If GDELT is not working by Friday it is cut to one chart, the reviewer signs off on Saturday, and we upload on Sunday well before 11:59 PM ET.

Methodology

We measure diversity with four simple metrics on a 20,000-user sample of MIND-small, and add one regional view from a one-month GDELT slice.

Data

- **MIND-small (train set):** `behaviors.tsv` gives each user's reading history and the articles they were shown (impressions) with click labels; `news.tsv` gives each article's category and subcategory.
- **GDELT:** one month of news coverage, counted by country, limited to the top ~50 countries so the chart stays readable.
- **Sampling:** a random sample of 20,000 MIND users with a fixed seed (42), stated on the methodology page so the result can be repeated.

Metrics

Metric	Question it answers	How to read it
Shannon entropy of categories (history, shown, clicked)	How varied is one user's news diet?	Higher = more varied; compare shown vs clicked
Gini coefficient of article exposure	Is exposure spread evenly across articles?	0 = equal, 1 = one article gets everything
Catalog coverage	How much of the catalog is ever shown?	Share of articles shown at least once
Popularity share	How much exposure goes to the top 10% of articles?	Higher = stronger popularity bias (to be added to the script)

Groups. Users are grouped by their most-read category in their history (at least 10 articles; groups of at least 50 users). These are behavior-based groups, not demographic communities.

GDELT view. Share of coverage by country, with the most- and least-covered countries side by side. Whether to compare it with population is still an open question (see the last section).

Data terms and citation. MIND is released free for research use under the Microsoft Research License Terms. We cite Wu et al., ACL 2020 (the MIND paper) and state that we used a Hugging Face copy whose counts match the official ones. GDELT describes its data as free and open.

Sample sizes. Every figure and chart in the PDF carries its sample size.

- Personalization findings use the 10,785 users (of 20,000 sampled) with at least 10 articles in their reading history.

- Group boosts are quoted as per-user medians; the heatmap shows each group's combined (pooled) feed, and its footnote says so. Small groups (movies 93 users, travel 89, music 54) are not used for headlines.
- Popularity concentration: the top 10% of the 16,841 articles shown at least once (1,685 articles) get 87% of impressions. Over the full 51,282-article catalog the figure is 98%, but that is inflated because about two-thirds of the catalog was never shown, so we report "of articles shown".
- GDELT: 4.45 million events across 247 countries and territories; 2.6% of rows have no country code and 0.06% have a code missing from the lookup, and both are dropped.

Data windows. MIND-small's train impressions cover Nov 9 to 14, 2019. The GDELT slice covers Oct 16 to Nov 14, 2019 (30 daily files ending on MIND's last day). The periods overlap but are not identical, so we compare patterns, not exact values.

Deliverable blueprint

The PDF has five pages, and each page answers one question so a judge can follow the story in a single pass.

Page	Purpose	Must show
1. Hook and question	Make the problem concrete in 10 seconds	One striking number from the data, the research question, the team name
2. Data and method	Prove the work is sound	Sources, sample size, the four metrics in plain words
3. Who sees what (MIND)	Answer exposure and personalization	Shown vs clicked by category; entropy of shown vs clicked; narrowest reader groups
4. Where news looks (GDELT)	Add the geographic gap	Coverage share by country, most vs least covered
5. Takeaways and limitations	Close the story and lead to Phase 2	Three findings, honest limitations, one line on the tool or framework we will build next

Design rules

- One message per chart, written as the chart title (for example, "Readers of one category see a narrower feed").
- One colour palette and one font across all five pages; large, readable numbers.
- Every chart names its data source and sample in small text underneath.
- Export to PDF early (Friday) to check layout, file size and that the file opens correctly.

Risks, limitations and open questions

The biggest risk is time: four days is enough for one clear story, not two deep analyses.

Risk or limitation	Fallback or how we state it
MIND is large and slow to process	Use MIND-small and the 20,000-user sample
GDELT is huge and messy	One month, top ~50 countries, one or two charts; if it stalls by Friday, cut it to one chart
No demographics in MIND	Use behavior-based groups and say so on page 5
Clicks are not the same as interest	State it as a limitation; avoid claims about what users "want"
Data is from one platform and one period (2019)	State that results show the pattern, not today's feeds
We measure shown vs clicked, not the real ranking algorithm	Describe findings as exposure patterns, not proof of cause

Also state on page 5: the MIND window (Nov 9 to 14, 2019) and the GDELT window (Oct 16 to Nov 14, 2019) overlap but are not identical, so the two views are compared as patterns, not as matching numbers.

Open questions for the team

- [ ] How many teammates do we have, and who takes which role?
- [ ] Do we compare GDELT coverage with population? (needs a second dataset)
- [ ] Which single number opens page 1?
- [ ] Which tool builds the final PDF (for example Canva, Figma or Google Slides)?